

Curriculum Vitae - Li Hongmin

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Primary Affiliation: Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo ([Laboratory](#))

Guest Affiliation: Frith Lab, The University of Tokyo ([Laboratory](#))

Research Interests

- AI-Automated Scientific Workflows
 - AI for Science and Scientific Discovery
 - AI Research and Evaluation
 - Bioinformatics and Biological Sequence Analysis
 - Biomolecular Sequence Design and Optimization
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Education

- 2019.04 — 2022.03 | Ph.D. in Computer Science (Information Systems Engineering) — University of Tsukuba
 - 2017.04 — 2019.03 | Master's in Computer Science (Information Systems Engineering) — University of Tsukuba
 - 2011.09 — 2015.07 | Bachelor's in Electronic Information Engineering — Ningxia University
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Academic & industry appointments

- 2026.6 — Present | Founder — AlphaScience Lab (Research community); [Website](#)
 - 2026.4 — Present | Researcher — Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo; [Laboratory](#)
 - 2026.4 — Present | Guest Researcher — Frith Lab, The University of Tokyo; [Laboratory](#)
 - 2023.5 — 2026.3 | Postdoctoral Researcher — The University of Tokyo
 - 2022.10 — 2023.5 | Machine Learning Engineer — HAOMO.AI — Autonomous-driving project · [蓝色空间领航者计划](#)
 - 2022.4 — 2022.10 | Postdoctoral Researcher — The University of Tokyo; [Details](#)
 - 2021.7 — 2021.9 | Machine Learning Engineer (Internship) — HAOMO.AI
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Research funding & computing support

- 2026.06 | Google Cloud TPU Builders Award — Cloud computing support · USD 7,500 in Google Cloud computing credits
 - 2026.04 | Grant-in-Aid for Early-Career Scientists (KAKENHI) — Development of an Input Data Differentiable Integrated Framework (ID3) for Biomolecular Sequence Design · ¥4,550,000; [Project info](#)
 - 2025.10 | Google research support — Biological sequence optimization · USD 30,000
 - 2024.4 | Grant-in-Aid for Early-Career Scientists (KAKENHI) — Development of a Large-Scale Language Model Integrating RNA Sequences and Text · ¥4,420,000; [Project info](#)
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Publications

Books

- 2026.08 | Hongmin Li. “Academic Collapse Does Not Happen All at Once: A Conditional Theory of Epistemic Cascades, Expertise Cliffs, and Institutional Recoupling”. Book · Independently published · 310 pp. · ISBN 979-8192323182. [Amazon](#)

Preprints

- 2026.09 | Linghu Fan, Hongmin Li. “Resolutions of Linear p -Cyclic Quotient Singularities”. arXiv preprint · 2609.07182 [math.AG]. [arXiv](#)
- 2026.06 | Hongmin Li. “The Calibration Turn in AI-Assisted Research: A Conceptual and Methodological Framework for Evidence-Licensed Claims”. arXiv preprint · 2606.31273 [cs.LG]. [arXiv](#), [Code & artifacts](#)
- 2026.05 | Hongmin Li. “Targeted Tests for LLM Reasoning: An Audit-Constrained Protocol”. arXiv preprint · 2605.11599 [cs.LG]. [arXiv](#)
- 2026.05 | Hongmin Li. “Separating Shortcut Transition from Cross-Family OOD Failure in a Minimal Model”. arXiv preprint · 2605.12945 [cs.LG]. [arXiv](#)
- 2026.05 | Hongmin Li. “A Controlled Counterexample to Strong Proxy-Based Explanations of OOD Performance: in a Fixed Pretraining-and-Probing Setup”. arXiv preprint · 2605.11554 [cs.LG]. [arXiv](#)
- 2026.05 | Hongmin Li. “FastUMAP: Scalable Dimensionality Reduction via Bipartite Landmark Sampling”. arXiv preprint · 2605.11428 [cs.LG]. [arXiv](#)
- 2025.12 | Bian Bian, Yiming Zhang, Hongmin Li, Jiuzhou Zhong, Yutaka Saito. “Large generative mRNA language foundation model for efficient coding sequence generation and design with mRNA-GPT”. bioRxiv. [Preprint](#)
- 2023.9 | Hao Wu, Xiangjun Pei, Takashi Oguchi, Hongmin Li, Zifei Wang, Zhihao He, Runqiu Huang. “An Integrated Physical Approach to Earthquake-Induced Landslide Susceptibility Incorporating Geological Structure: A Case Study of the Diexi Catchment, Sichuan, China”. SSRN preprint. [SSRN](#)

Journal Articles

- 2026.09 | Hongmin Li, Goro Terai, Takumi Otagaki, Kiyoshi Asai. “Gradient-based Optimization for mRNA Sequence Design”. Bioinformatics. [Journal / DOI](#), [Preprint](#), [Code](#)
- 2023.1 | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. “LSEC: Large-scale spectral ensemble clustering”. Intelligent Data Analysis. [arXiv](#), [Code](#)
- 2022.8 | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. “Divide-and-conquer based Large-Scale Spectral Clustering”. Neurocomputing. [Preprint](#), [Code](#)
- 2022.7 | Meng Huang, Xiucan Ye, Hongmin Li, Tetsuya Sakurai. “Missing Value Imputation With Low-Rank Matrix Completion in Single-Cell RNA-Seq Data by Considering Cell Heterogeneity”. Frontiers in Genetics. [DOI](#)
- 2020.7 | Xiucan Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. “An Oversampling Framework for Imbalanced Classification Based on Laplacian Eigenmaps”. Neurocomputing. [Link](#)
- 2019 | Xiucan Ye, Hongmin Li, Tetsuya Sakurai, Pei-Wei Shueng. “Ensemble Feature Learning to Identify Risk Factors for Predicting Secondary Cancer”. International Journal of Medical Sciences. [DOI](#)

Conference Papers

- 2020.11 | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. “Ensemble Learning for Spectral Clustering”. ICDM 2020. [PDF](#), [Code](#)

- 2020.7 | Hongmin Li, Xiucan Ye, Akira Imakura, Tetsuya Sakurai. "Hubness-based Sampling Method for Nyström Spectral Clustering". IJCNN 2020. [Link](#)
 - 2019.8 | Xiucan Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. "Distributed Collaborative Feature Selection Based on Intermediate Representation". IJCAI 2019. [Link](#)
 - 2018.10 | Xiucan Ye, Hongmin Li, Tetsuya Sakurai, Zhi Liu. "Large Scale Spectral Clustering Using Sparse Representation Based on Hubness". CBDCOM 2018. [Link](#), [Code](#)
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Conference presentations

- 2026.06.09 | Poster presentation — JST CREST BioDX Area Meeting — 5th area meeting · Numazu, Japan
 - 2025.11 | Poster presentation — JST CREST BioDX Interim Symposium — [Event](#), [CREST 2025 poster](#)
 - 2025.8 | Oral presentation — Input Data Differentiable Designer (ID3) at RNA Informatics Dojo 2025 — Early presentation of the method later developed into the preprint "Gradient-based Optimization for mRNA Sequence Design"; [Event](#), [Paper](#), [Code](#)
 - 2024.10 | Poster presentation — FastUMAP at the Asia-Pacific Bioinformatics Joint Conference 2024 — Early conference presentation of the work later developed into the FastUMAP preprint; [Event](#), [Paper](#)
 - 2024.8 | Oral presentation — RNA Informatics Dojo 2024
 - 2024.8 | Participation — IJCAI 24 (Jeju)
 - 2024.6 | Participation — WCCI 24 (Yokohama)
 - 2024.6 | Participation — 25th Annual Meeting of the RNA Society of Japan
 - 2023.8 | Oral presentation — RNA Informatics Dojo 2023
 - 2022.8 | Oral presentation — RNA Informatics Dojo 2022
 - 2022.3 | Presentation — JST SPRING recipients event
 - 2021.12 | Oral presentation — CollaboTICS 2021
 - 2019.6 | Poster presentation — G20 Digital Science Now Symposium
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Peer Review Activities

- 2026.01 | The Journal of Supercomputing
 - 2024.11 | Pattern Analysis and Applications
 - 2024.11 | Knowledge and Information Systems
 - 2024.10 | Neurocomputing
 - 2024.09 | The Visual Computer
 - 2024.03 | International Journal of Machine Learning and Cybernetics
 - 2022.11 | Transactions on Pattern Analysis and Machine Intelligence
 - 2022.10 | Briefings in Functional Genomics
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Honors & fellowships

- 2021.10 | Doctoral research support — JST SPRING — Support for Pioneering Research Initiated by the Next Generation
 - 2020.10 | AETA Earthquake Prediction AI Algorithm Competition 2019 — Second Prize — Second-place recognition for an AI model predicting earthquakes from real-time monitoring data · awarded in 2020; [News](#), [Slides](#)
 - 2019.10 | 3rd Analysys International Algorithm Competition — Special Award — Special recognition in the website page-view and unique-visitor prediction task
 - 2018.10 | IEEE International Conference on Cloud and Big Data Computing 2018 — Best Paper Award — Awarded for "Large Scale Spectral Clustering Using Sparse Representation Based on Hubness"
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Skills

Programming Languages

- Python, R, MATLAB, C/C++, Java

Machine Learning & Deep Learning

- PyTorch, TensorFlow, scikit-learn
- Spectral Clustering, Large Language Models
- Bioinformatics Applications

Computational Biology

- RNA Sequence Analysis
- mRNA Design and Optimization
- Single-cell RNA-seq Data Analysis